MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 26: Examples of conformal mappings

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Lecture outline

1. Elementary transformations
2. Examples
3. Linear mapping as a sequence of elementary transformations
4. Examples

Example 1

Find magnification and inclination angle of $w = z^2$ at $a = 1 + i$. Solution.

First we find $w'(z) = 2z$ and $w'(1+i) = 2 + 2i$. Therefore, the magnification is

$$|w'(1+i)| = |2+2i| = \sqrt{2^2 + 2^2} = 2\sqrt{2}.$$

To find the inclination we consider:

$$arg\, w'(1+i) = arg\,(2+2i) = \frac{\pi}{4}$$

All smooth curves going through the point $i + 1$ will have the inclination $\pi/4$ and the magnification $2\sqrt{2}$, meaning that their images at $b = (1+i)^2 = 2i$ turns the same angle $\pi/4$ and magnify $2\sqrt{2}$ times.

Example 2. Part 1

Consider a function $w = z^2 - 2z$. Find a) points where this function is not conformal, and b) magnification and inclination angle at points 1) $a = 5$, 2) $a = 1 + i$, 3) $a = -i$.

Solution.

a) The derivative of this function is

$$w'(z) = 2z - 2 = 2(z - 1).$$

The derivative $w'(z) = 0$ at $z = 1$, and this is the only point where this mapping is not conformal.

Example 2. Part 2

b) Consider given points:

1. Let $a = 5$ then $w'(5) = 8$, and $|w'(5)| = 8$, and $arg\ w'(5) = 0$. Therefore, all smooth curves passing the point $b = 5^2 - 2 \cdot 5 = 15$ do not turn (inclination angle is zero) and magnify in 8 times.
2. Let $a = 1 + i$ then $w'(1 + i) = 2i$, and $|w'(1 + i)| = |2i| = 2$, and $arg\ w'(1 + i) = \pi/2$. Therefore, images of all smooth curves passing through $1 + i$ turn $\pi/2$ and magnify in 2 times.
3. Let $a = -i$ then $w'(-i) = 2(-1 - i)$, and $|w'(-i)| = 2\sqrt{2}$, and $arg\ w'(-i) = -3\pi/4$. Therefore, images of all smooth curves passing through $-i$ turn in the negative direction $-3\pi/4$ and magnify in $2\sqrt{2}$ times.

Elementary mappings

It is useful for the problems solving to consider the following elementary maps:

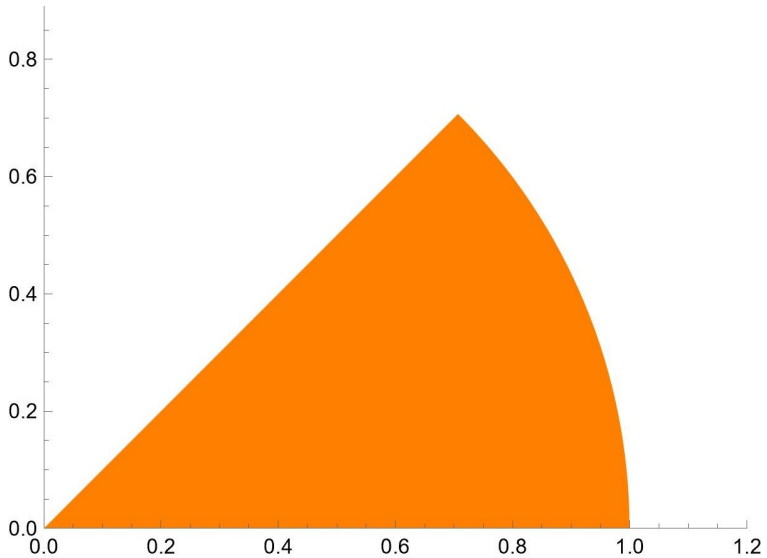
1. Parallel transfer: $w = z + c$, where c is a constant
2. Rotation around the origin: $w = ze^{i\alpha}$, where α is a real number. $\alpha > 0$ - positive direction, $\alpha < 0$ - negative direction
3. Scaling: $w = rz$, where $r > 0$, $r < 1$ - shrinking, $r > 1$ - magnifying.
4. Inversion: $w = 1/z$

Example 3. Part 1

Find an image of the set E

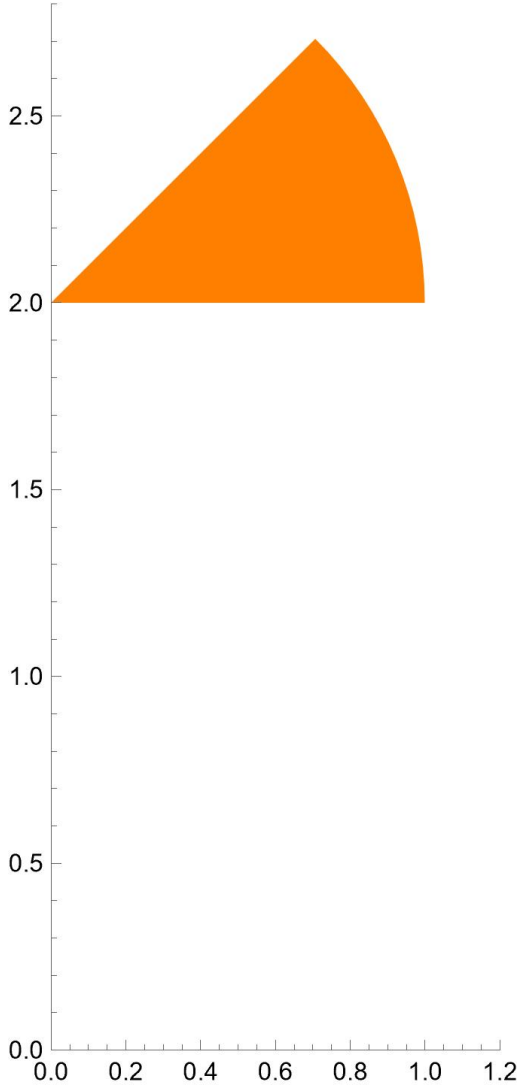
$$E = \{z : |z| < 1, 0 < argz < \pi/4\}$$

when the following mappings are applied:
a) $w = z + 2i$,
b) $w = ze^{i\pi/4}$,
c) $w = 2z$



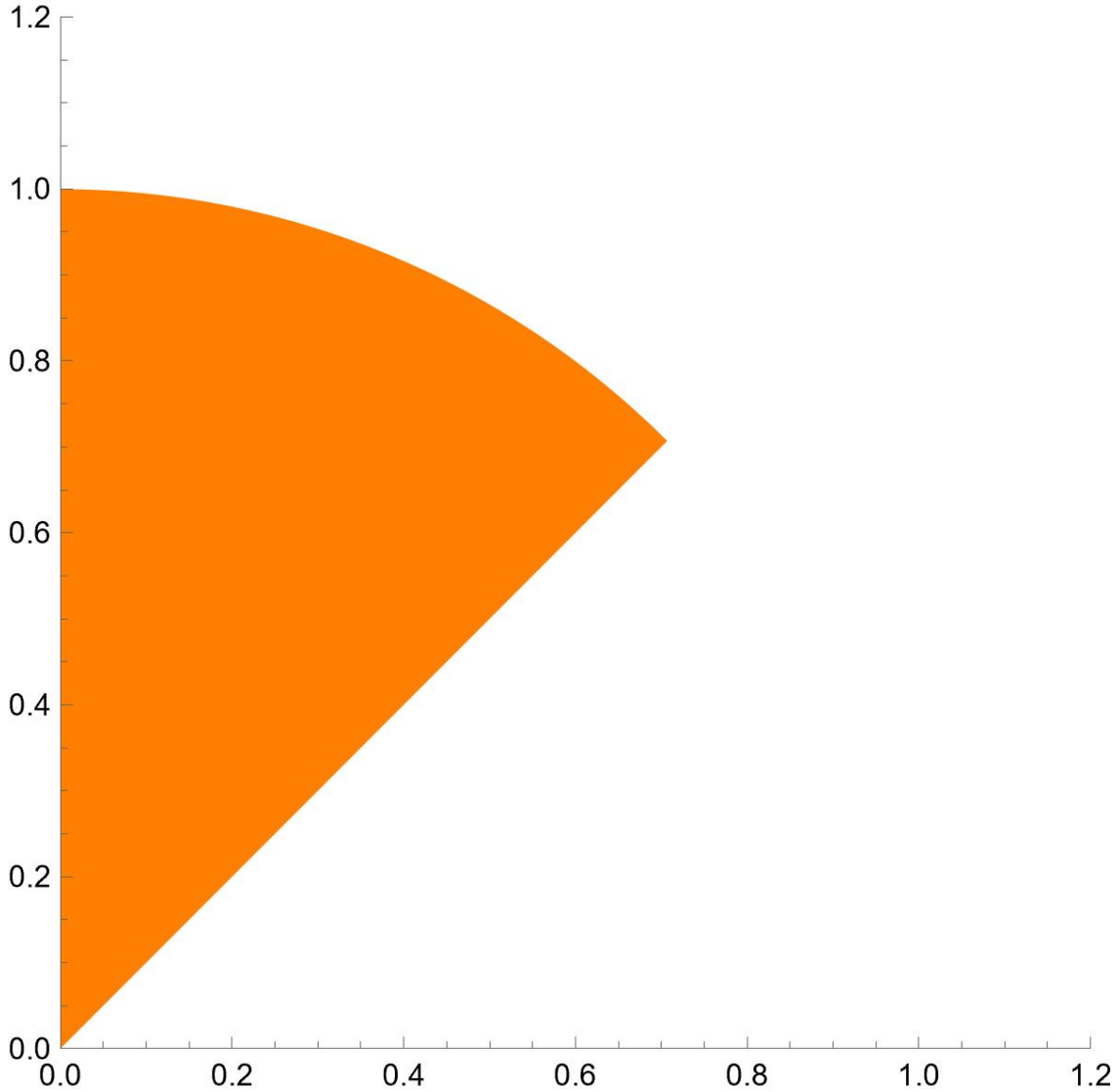
Example 3. Part 2

a) $w = z + 2i$ transfers E up 2 units



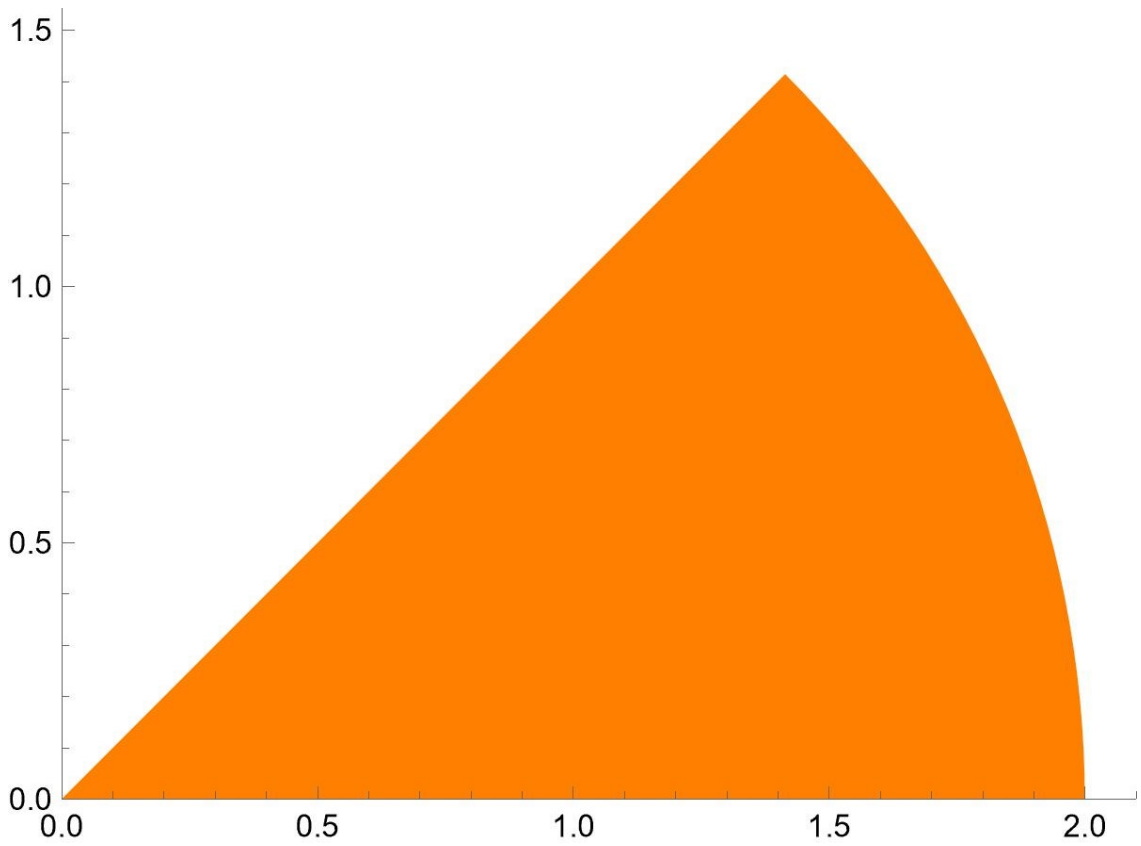
Example 3. Part 3

b) $w = ze^{i\pi/4}$ rotates E in the positive direction on $\pi/4$



Example 3. Part 4

c) $w = 2z$ magnifies E two times



Example 4.

Find how the function $w = 1/z$ maps the generalized circle (which includes straight line)

$$a|z|^2 + \bar{b}z + b\bar{z} + c = 0$$

where $ac < |b|^2$, a and c are real numbers, b - complex number.

Substitute $z = 1/w$ in to the equation to get:

$$a\frac{1}{|w|^2} + \bar{b}\frac{1}{w} + b\frac{1}{\bar{w}} + c = 0$$

multiplication by $w\bar{w}$ we get:

$$a + \bar{b}\bar{w} + bw + c|w|^2 = 0$$

Which is also an equation of the generalized circle.

Example 5.

Find how the function $w = 1/z$ maps the following set:

$$E = \{z : Im\, z = 0\}$$

The set E is a real axis. Let us find $z(w)$: from $w = 1/z$ it follows that $z = 1/w$. Substitute $z = 1/w$ into E: $Im\, z = 0$ then $Im\, 1/w = 0$. We can use the fact that $w\bar{w} = |w|^2$, then

$$Im\left(\frac{1}{w}\right) = Im\left(\frac{\bar{w}}{w\bar{w}}\right) = Im\left(\frac{\bar{w}}{|w|^2}\right) = \frac{1}{|w|^2} Im\, \bar{w}$$

Therefore, $Im\, \bar{w} = 0$ or $Im\, w = 0$ and the image of the real axis is the real axis again.

Linear function in terms of elementary maps

Let us consider the linear function:

$$f(z) = az + b$$

, where a,b are complex numbers. This function is regular on the entire plane

$w'(z) = a \neq 0$, and, therefore, it is a conformal map in all points.
Let us rewrite this function in the following way:

$$w = |a|e^{i\ arg\ a} \cdot z + b$$

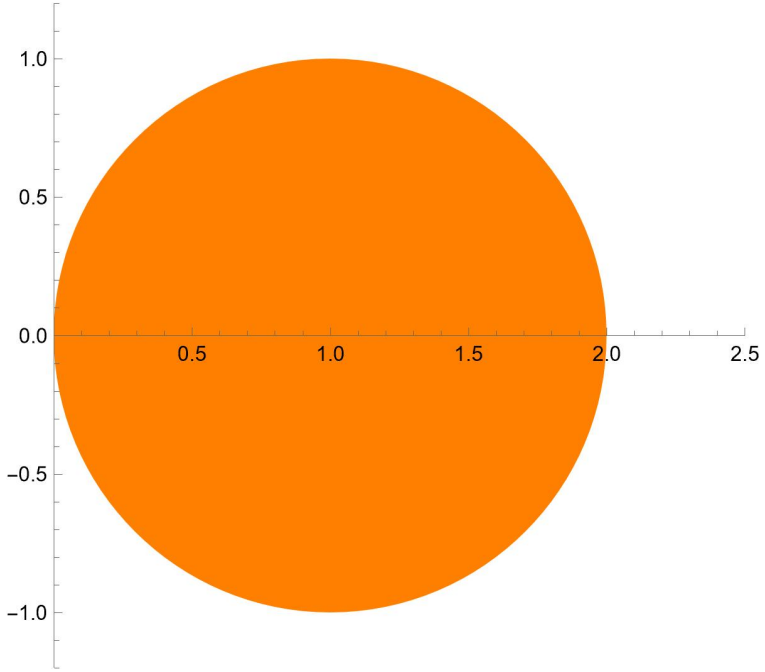
Then we can split this map into the combination of the following transformations:

1. $w_1 = e^{i\ arg\ a} z$ - turn on the angle $arg\ a$
2. $w_2 = w_1|a|$ - scaling by $|a|$
3. $w_3 = w = w_2 + b$ - transfer by b

Example 6. Part 1.

Find how the function $w = 1 - 2i\,z$ maps the following set:

$$E = \{z : |z - 1| < 1\}$$

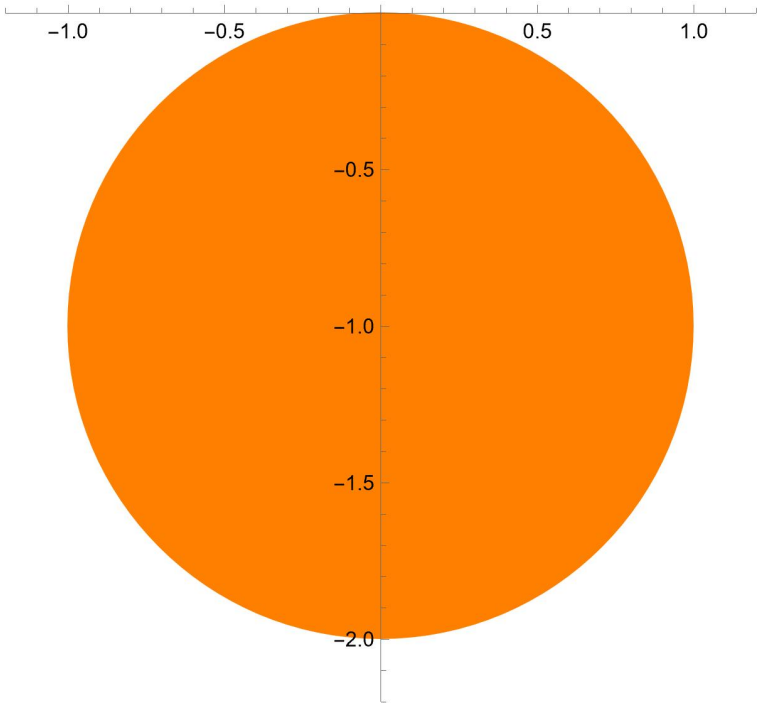


This function can be presented as the following sequence:

1. $w_1 = e^{-i\pi/2} z$ as soon as $-i = e^{-i\pi/2}$
2. $w_2 = 2w_1$ - scaling by 2
3. $w_3 = w = w_2 + 1$ - transfer by 1

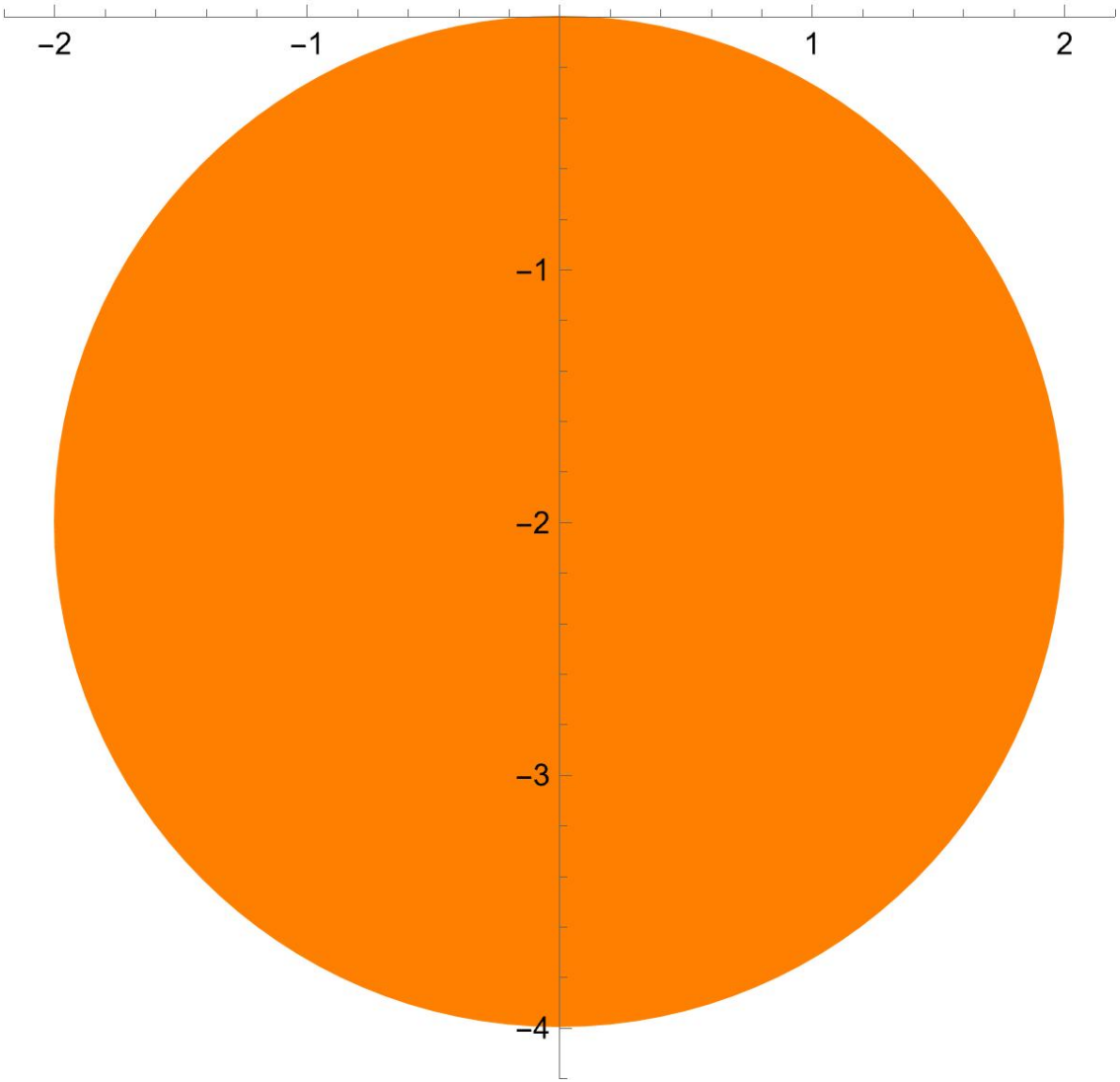
Example 6. Part 2.

First map, $w_1 = e^{-i\pi/2}z$, rotates the disk E on $-\pi/2$ radians



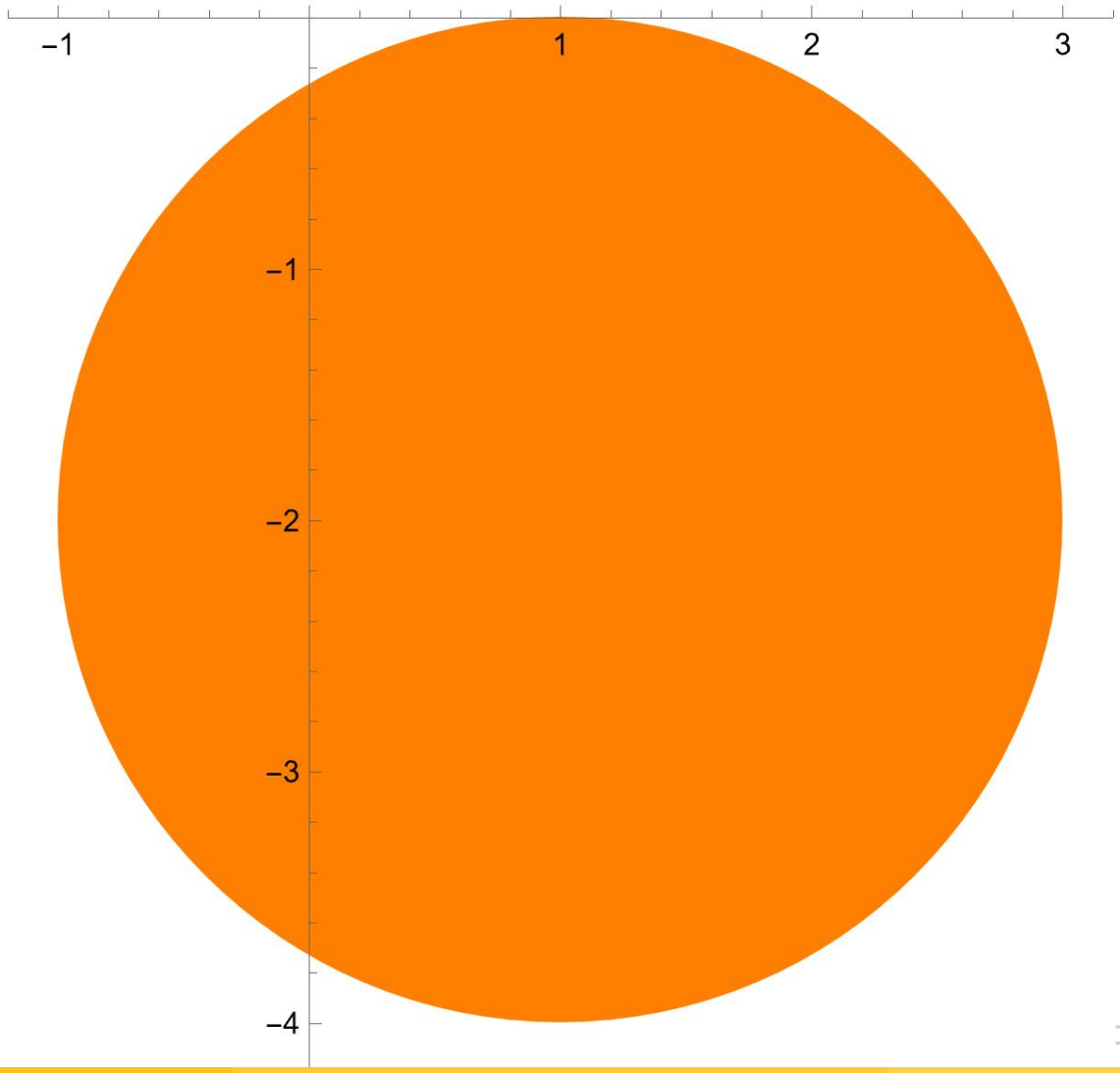
Example 6. Part 2.

The second map, $w_2 = 2w_1$, magnifies rotated disk E two times



Example 6. Part 3.

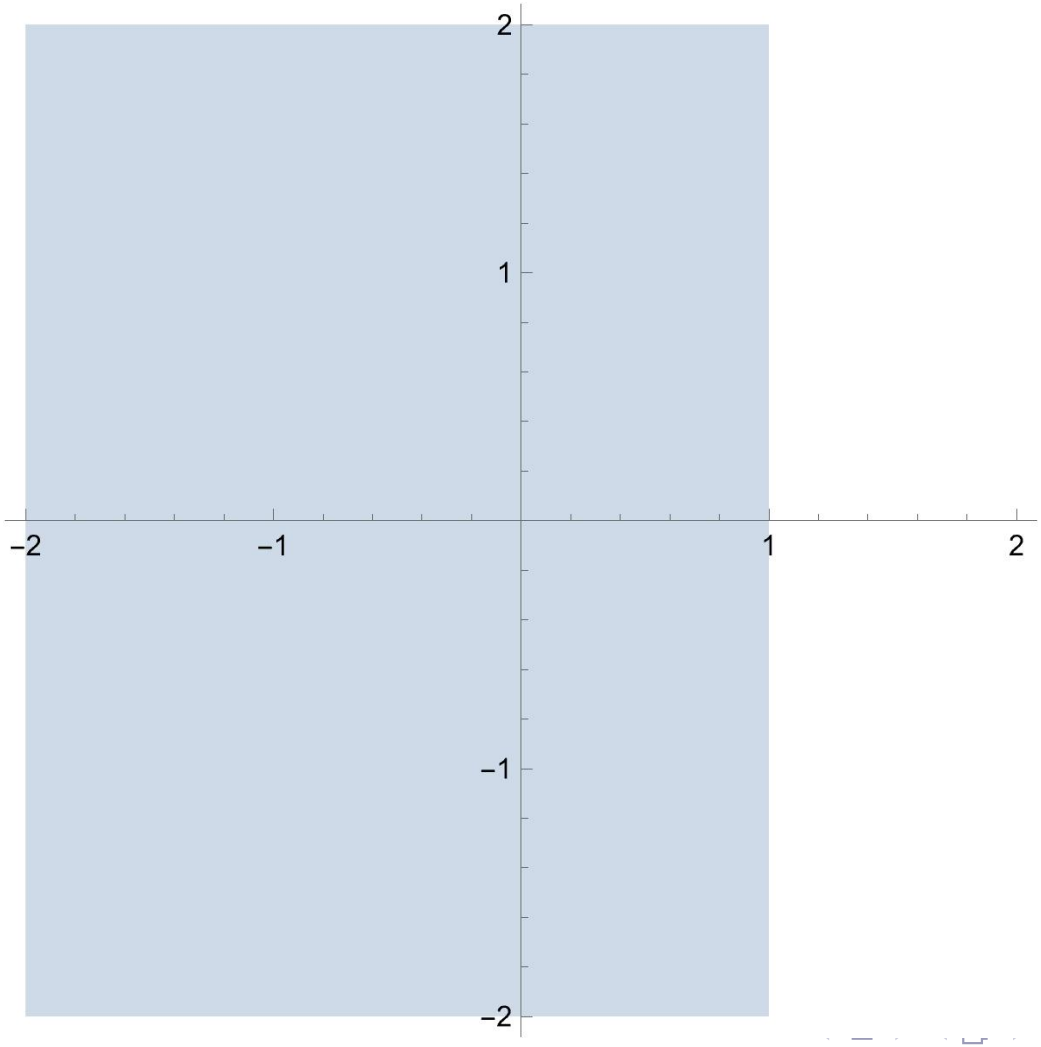
Finally, the third map, $w_3 = w_2 + 1$, moves magnified and rotated disk E on one unit right. The answer is $w(E) = \{w : |w - 1 + 2i| < 2\}$.



Example 7. Part 1.

Find how the function $w = (1+i)\,z + 1$ maps the following set:

$$E = \{z : Re\, z < 1\}$$

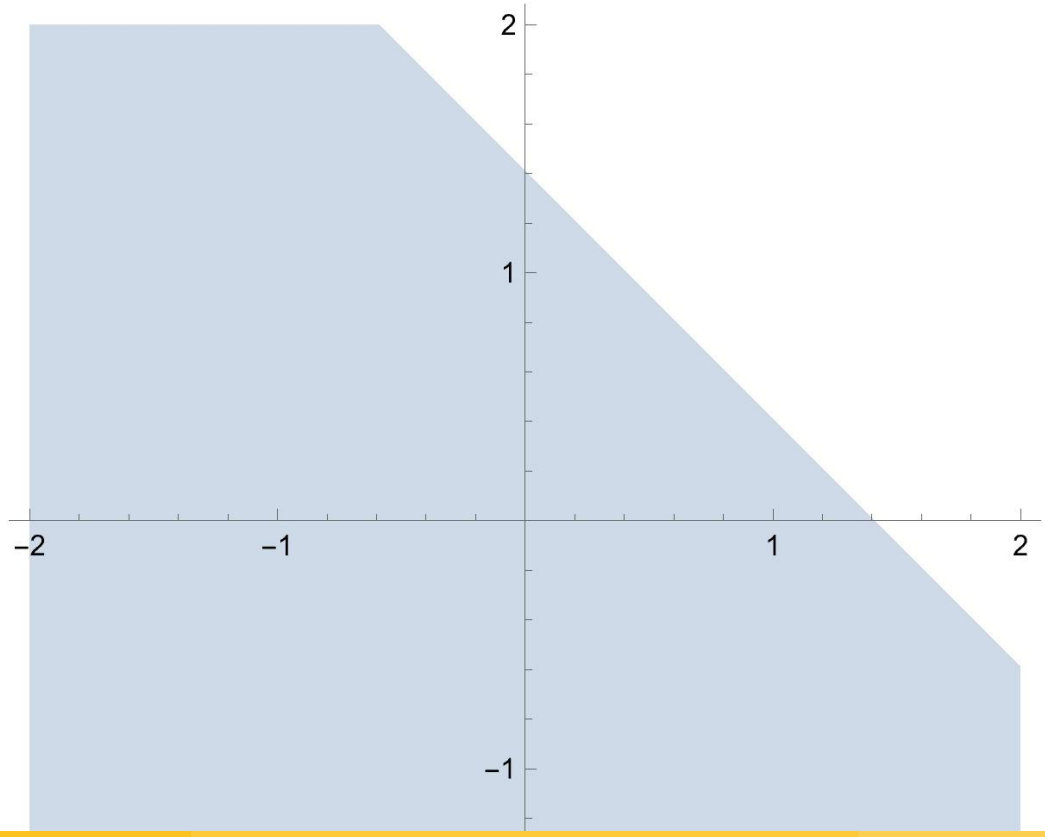


Example 7. Part 2.

This function, $w = (1+i)\,z + 1$, can be presented as the following sequence:

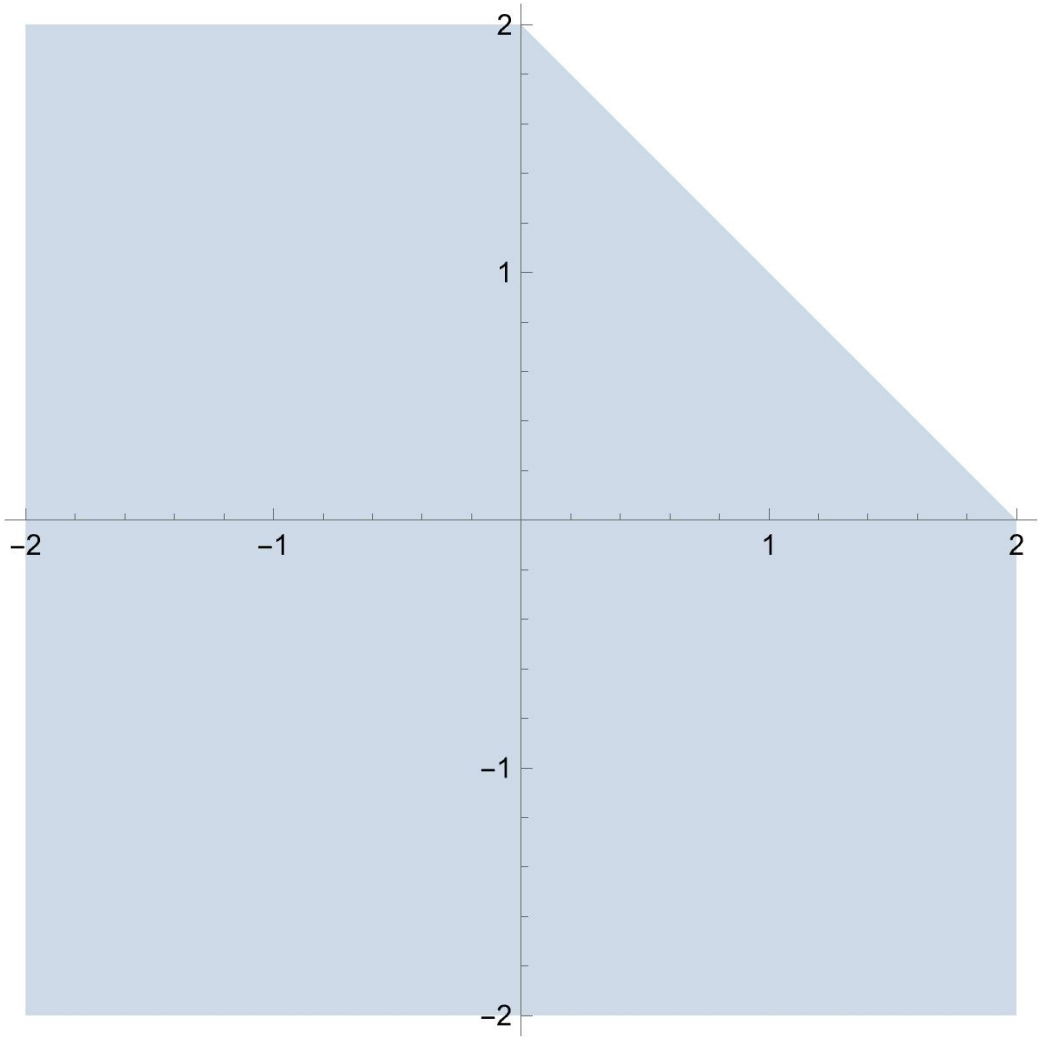
1. $w_1 = e^{i\pi/4} z$ as soon as $1 + i = \sqrt{2} e^{i\pi/4}$
2. $w_2 = \sqrt{2} w_1$
3. $w_3 = w_2 + 1$

The first step is rotation by $\pi/4$:



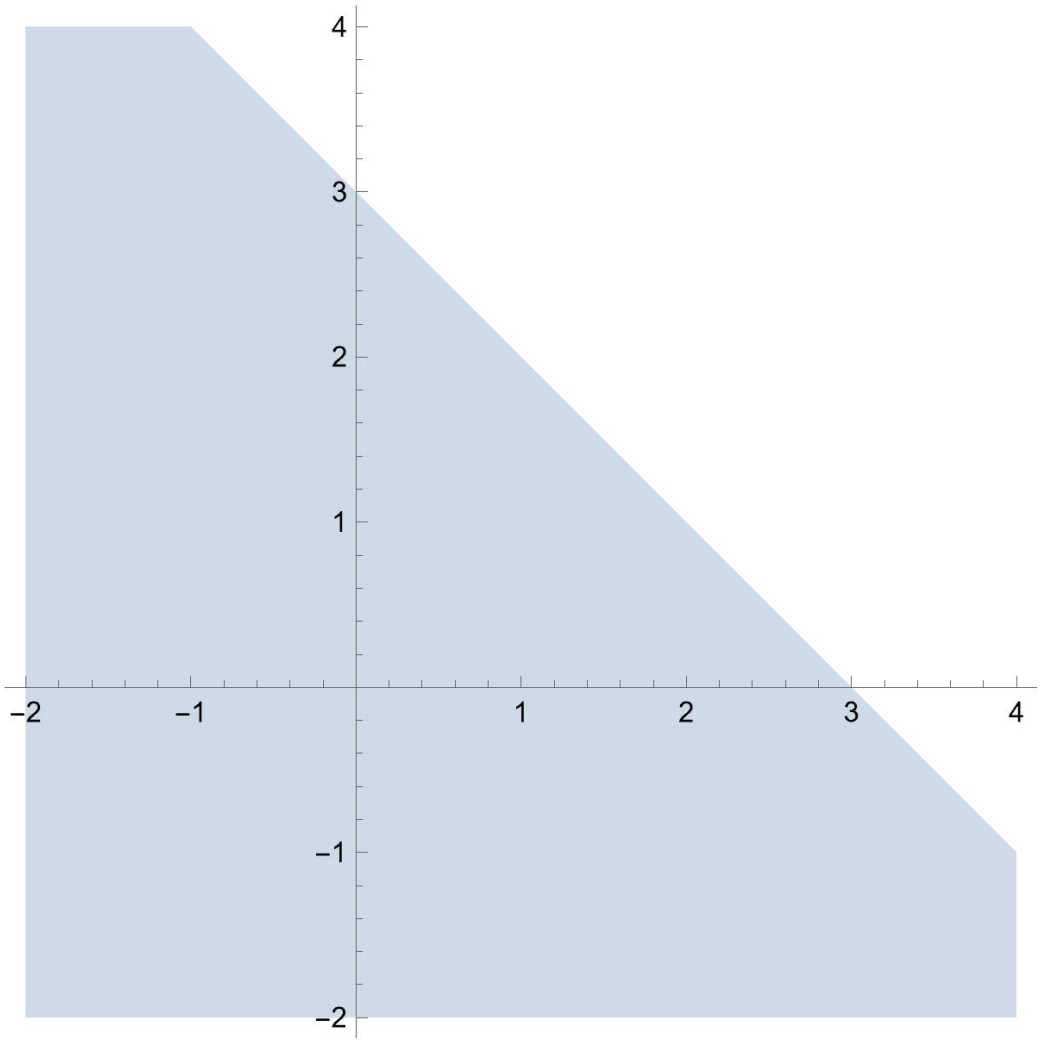
Example 7. Part 3.

The second step is scaling by $\sqrt{2}$:



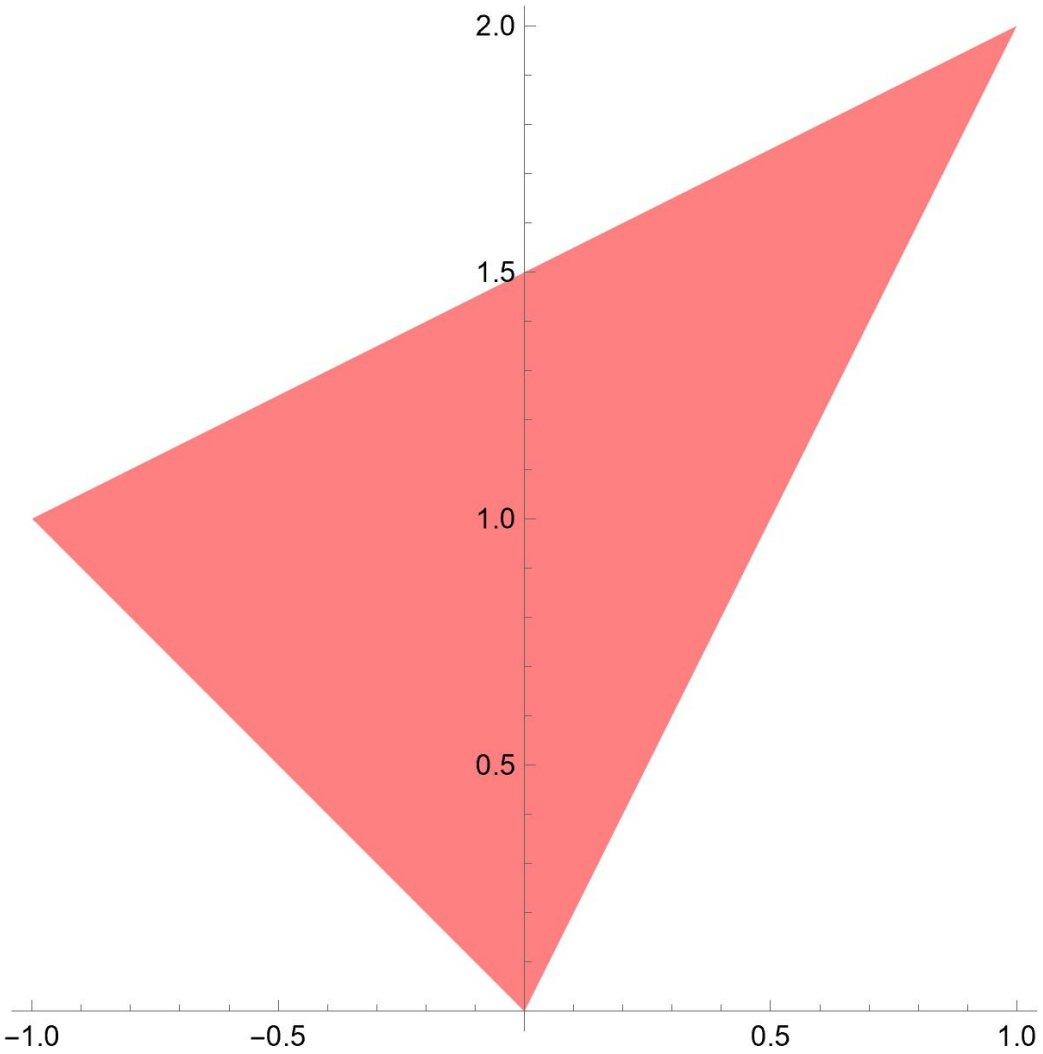
Example 7. Part 4.

The last step is shifting by 1. The answer is $w(E) = \{w : Re\, w + Im\, w < 3\}$.



Example 8. Part 1.

Let E be a triangle with vertices at $z_1 = 0$, $z_2 = -1 + i$, $z_3 = 1 + 2i$, Find the image of this set with the map $w = (4 + 3i)z + 3 - 2i$.



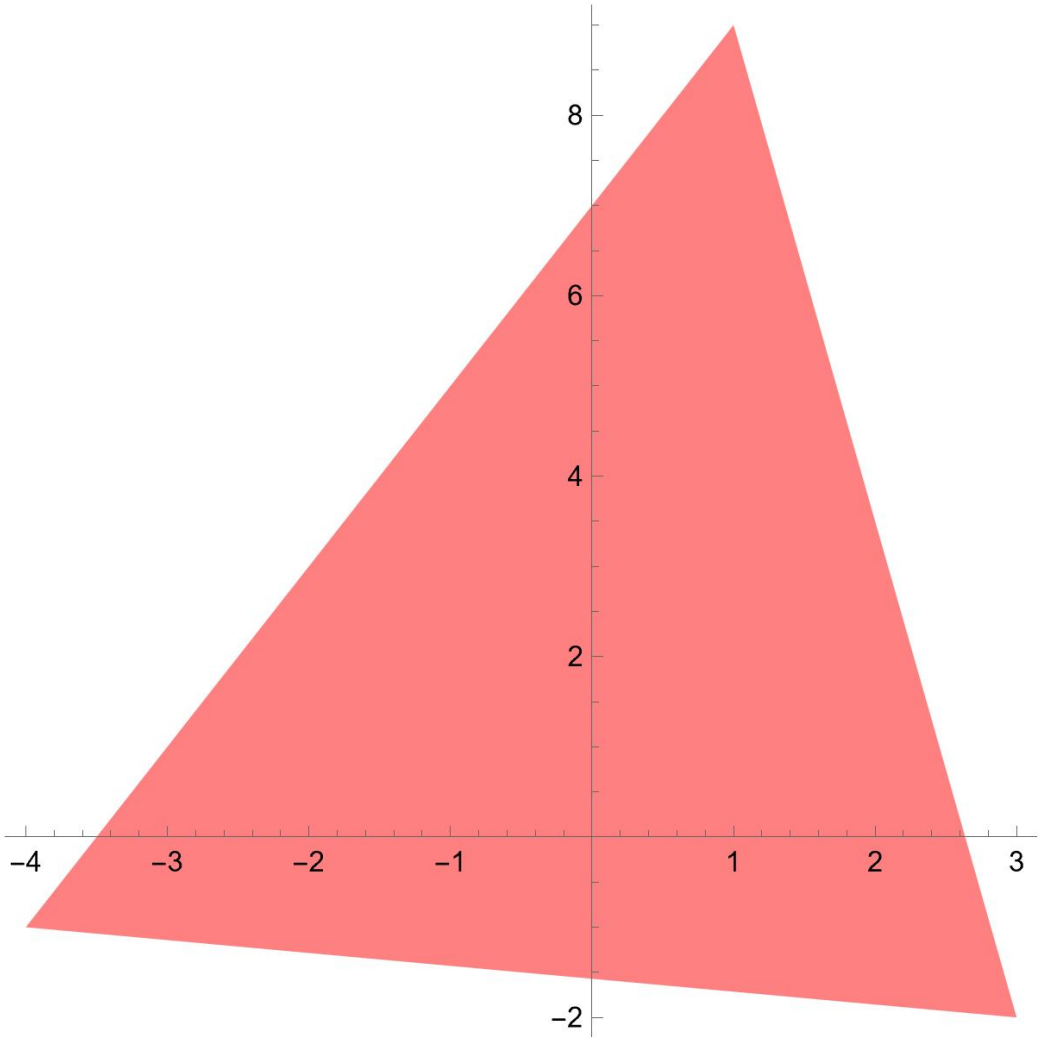
Example 8. Part 2.

The map $w = (4+3i)z + 3 - 2i$ is conformal on the whole plane, then, all angles of the triangle will be preserved. Therefore, the image is a similar triangle rotated, scaled and shifted. The function can be rewritten as

$$w = (4+3i)z + 3 - 2i = 5e^{iarctan3/4}z + (3-2i)$$

Then the image of the triangle is the result of rotation by $arctan3/4$, magnifying it in 5 times and transfer by $3-2i$. The coordinates of the vertices are : $w(z_1) = 3-2i$, $w(z_2) = -4-i$, $w(z_3) = 1+9i$.

Example 8. Part 3.



Thank you